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Reverse isomerization process using feedstock containing dissolved hydrogen

Abstract

A process for reverse isomerization of a light naphtha feedstock containing branched C.sub.5-C.sub.7 paraffins. The process includes feeding a mixed feed stream including the light naphtha feedstock to a separation unit to generate an iso-paraffin stream and one or more normal paraffin streams. The process further includes mixing hydrogen gas and a hydrocarbon feed stream containing the iso-paraffin stream to form a hydrogen-enriched liquid feed stream which is provided to a reverse isomerization reactor. The hydrogen-enriched liquid feed stream is contacted with a solid reverse isomerization catalyst for reverse hydroisomerization in a substantially two-phase liquid-solid reverse isomerization fixed-bed reaction zone convert iso-paraffins to normal paraffins. The isomerization effluent stream is provided to a stabilization column to generate a stabilized isomerate stream which is combined with the light naphtha feedstock to generate the mixed feed stream. An associated system for performing the process is also provided.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation application of U.S. patent application Ser. No. 18/055,531, filed Nov. 15, 2022, the entire contents of which are hereby incorporated by reference in the present disclosure.

TECHNICAL FIELD

(1) The present disclosure relates to processes for reverse isomerization of a light naphtha feedstock comprising branched C.sub.5-C.sub.7 paraffins and more particularly reverse isomerization processes using a hydrogen-enriched liquid feed stream.

BACKGROUND

(2) Ethylene and propylene are important chemicals for use in the production of other useful materials, such as polyethylene and polypropylene. Polyethylene and polypropylene are two of the most common plastics found in use today and have a wide variety of uses. Uses for ethylene and propylene include the production of vinyl chloride, ethylene oxide, ethylbenzene and alcohol.

(3) Typically ethylene utilized for the production of plastics and petrochemicals such as polyethylene is generated by the thermal cracking of low molecular weight hydrocarbons. One such

method involves mixing steam with a feed stream to the cracking reactor to reduce the hydrocarbon partial pressure and enhance olefin yield and to reduce the formation and deposition of carbonaceous material in the cracking reactors. The process is therefore often referred to as a steam cracking reaction or pyrolysis.

(4) The composition of the feed to the steam cracking reactor affects the product distribution. As one skilled in the art understands, certain hydrocarbons are easier to crack into shorter chain molecules for generation of ethylene than other hydrocarbons. Typically normal paraffins are the easiest to crack followed by iso-paraffins, olefins, naphthenes, and aromatics each being sequentially more challenging. As such, increasing the concentration of normal paraffins in a stream can improve the quality of a feedstock to the steam cracking unit.

SUMMARY

(5) Accordingly, an efficient process for increasing the concentration of normal paraffins in the feedstream to a steam cracking reactor would yield a significant increase in the profitability of steam cracking operations by increasing the yield of high value products such as ethylene and propylene.

(6) There is a clear and long-standing need to provide an efficient and economical process for the production of ethylene and propylene, from a feedstock comprising substantial quantities of light naphtha and in particular shifting of the balance of normal paraffins and iso-paraffins to capture increased quantities of normal paraffins.

(7) In accordance with one or more embodiments of the present disclosure, a process for reverse isomerization of a light naphtha feedstock comprising branched C.sub.5-C.sub.7 paraffins is disclosed. The process includes feeding a mixed feed stream comprising the light naphtha feedstock to a separation unit to generate an iso-paraffin stream and one or more normal paraffin streams; mixing hydrogen gas and a hydrocarbon feed stream comprising the iso-paraffin stream to form a hydrogen-enriched liquid feed stream; providing the hydrogen-enriched liquid feed stream to a reverse isomerization reactor; operating the reverse isomerization reactor by contacting the hydrogen-enriched liquid feed stream with a solid reverse isomerization catalyst for reverse hydroisomerization in a substantially two-phase liquid-solid reverse isomerization fixed-bed reaction zone under conditions that minimize cracking reactions and that are effective to convert iso-paraffins to normal paraffins and recovering an equilibrium composition of normal and iso paraffins in an isomerization effluent stream; providing the isomerization effluent stream to a stabilization column to separate the isomerization effluent stream into a C.sub.1-C.sub.4 hydrocarbon stream and a stabilized isomerate stream; and combining the stabilized isomerate stream with the light naphtha feedstock to generate the mixed feed stream.

(8) Additional features and advantages of the described embodiments will be set forth in the detailed description that follows. The additional features and advantages of the described embodiments will be, in part, readily apparent to those skilled in the art from that description or recognized by practicing the described embodiments, including the detailed description that follows as well as the drawings and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following detailed description of specific embodiments of the present disclosure can be best understood when read in conjunction with the following drawings in which:

(2) FIG. 1 is a schematic illustration of one or more embodiments of the process for reverse isomerization of a light naphtha feedstock of the present disclosure;

(3) FIG. 2 is a schematic illustration of one or more embodiments of the process for reverse isomerization of a light naphtha feedstock of the present disclosure as illustrated in FIG. 1

including a second light naphtha feed;

(4) FIG. 3 is a schematic illustration of one or more embodiments of the process for reverse isomerization of a light naphtha feedstock of the present disclosure including adsorption based separation;

(5) FIG. 4 is a schematic illustration of one or more embodiments of the process for reverse isomerization of a light naphtha feedstock of the present disclosure as illustrated in FIG. 3 including a second light naphtha feed;

(6) FIG. 5 is a schematic illustration of details of a mixing zone according to one or more embodiments of the present disclosure;

(7) FIG. 6 is a schematic illustration of details of the mixing zone of FIG. 5 including a flashing zone;

(8) FIG. 7 is a schematic illustration of details of a mixing zone according to one or more embodiments of the present disclosure;

(9) FIG. 8 is a schematic illustration of details of the mixing zone of FIG. 8 including a flashing zone.

(10) For the purpose of describing the simplified schematic illustrations and descriptions of FIGS. 1 through 8 the numerous valves, temperature sensors, electronic controllers, and the like that may be employed and well known to those of ordinary skill in the art of certain chemical processing operations are not included. Further, accompanying components that are often included in chemical processing operations, such as, for example, air supplies, heat exchangers, surge tanks, catalyst hoppers, or other related systems are not depicted. It would be known that these components are within the spirit and scope of the present embodiments disclosed. However, operational components, such as those described in the present disclosure, may be added to the embodiments described in this disclosure.

(11) It should further be noted that arrows in the drawings refer to process streams. However, the arrows may equivalently refer to transfer lines that may serve to transfer process streams between two or more system components. Additionally, arrows that connect to system components define inlets or outlets in each given system component. The arrow direction corresponds generally with the major direction of movement of the materials of the stream contained within the physical transfer line signified by the arrow. Furthermore, arrows that do not connect two or more system components signify a product stream which exits the depicted system or a system inlet stream which enters the depicted system. Product streams may be further processed in accompanying chemical processing systems or may be commercialized as end products. System inlet streams may be streams transferred from accompanying chemical processing systems or may be non-processed feedstock streams. Some arrows may represent recycle streams, which are effluent streams of system components that are recycled back into the system. However, it should be understood that any represented recycle stream, in some embodiments, may be replaced by a system inlet stream of the same material, and that a portion of a recycle stream may exit the system as a system product.

(12) Additionally, arrows in the drawings may schematically depict process steps of transporting a stream from one system component to another system component. For example, an arrow from one system component pointing to another system component may represent “passing” a system component effluent to another system component, which may include the contents of a process stream “exiting” or being “removed” from one system component and “introducing” the contents of that product stream to another system component.

(13) It should be understood that two or more process streams are “mixed” or “combined” when two or more lines intersect in the schematic flow diagrams. Mixing or combining may also include mixing by directly introducing both streams into a like reactor, separation device, or other system component. For example, it should be understood that when two streams are depicted as being combined directly prior to entering a separation unit or reactor, that in some embodiments the streams could equivalently be introduced into the separation unit or reactor and be mixed in the

reactor.

(14) Reference will now be made in greater detail to various embodiments, some embodiments of which are illustrated in the accompanying drawings. Whenever possible, the same reference numerals will be used throughout the drawings to refer to the same or similar parts.

DETAILED DESCRIPTION

(15) Reference will now be made in detail to embodiments of a process for reverse isomerization of a light naphtha feedstock to generate increased normal paraffin production in accordance with the present disclosure. While the associated system for reverse isomerization of a light naphtha feedstock comprising branched C.sub.5-C.sub.7 paraffins of FIGS. 1 through 8 are provided as exemplary embodiments, it should be understood that the present processes and systems may encompass other configurations.

(16) In one or more embodiments and generally with reference to FIG. 1 through 4, a process for reverse isomerization of a light naphtha feedstock **100** comprising branched C.sub.5-C.sub.7 paraffins includes feeding a mixed feed stream **110** comprising the light naphtha feedstock **100** to a separation unit **30** to generate an iso-paraffin stream **120** and one or more normal paraffin streams **130**. The process further includes mixing hydrogen gas **140** and a hydrocarbon feed stream **150** comprising the iso-paraffin stream **120** to form a hydrogen-enriched liquid feed stream **160**. Further, the process includes providing the hydrogen-enriched liquid feed stream **160** to a reverse isomerization reactor **10** and operating the reverse isomerization reactor **10** by contacting the hydrogen-enriched liquid feed stream **160** with a solid reverse isomerization catalyst **12** for reverse hydroisomerization in a substantially two-phase liquid-solid reverse isomerization fixed-bed reaction zone under conditions that minimize cracking reactions and that are effective to convert iso-paraffins to normal paraffins. An equilibrium composition of normal and iso paraffins is recovered in an isomerization effluent stream **170**. The isomerization effluent stream **170** is provided to a stabilization column **20** to separate the isomerization effluent stream **170** into a C.sub.1-C.sub.4 hydrocarbon stream **180** and a stabilized isomerate stream **190**. The stabilized isomerate stream **190** is then combined with the light naphtha feedstock **100** to generate the mixed feed stream **110** to generate a circular process.

(17) In one or more further embodiments and generally with reference to FIGS. 1 and 2, the separation unit **30** includes a deisopentanizer column **40** and a fractionator **50**. In accordance with such embodiments, the mixed feed stream **110** is provided to the deisopentanizer column **40** to separate iso-pentane as an iso-pentane stream **122** from the mixed feed stream **110** to generate a deisopentanizer effluent **200** comprising normal pentane and hexanes. The deisopentanizer effluent **200** is provided to the fractionator **50** to generate an iso-hexanes stream **124**, a normal pentane stream **132**, and a normal hexane stream **134**. The iso-pentane stream **122** and the iso-hexanes stream **124** are then combined to generate the iso-paraffin stream **120**.

(18) In one or more further embodiments and generally with reference to FIGS. 3 and 4, the separation unit **30** includes a molecular sieve separation column **60**. The molecular sieve separation column **60** separates the mixed feed stream **110** into the iso-paraffin stream **120** and the one or more normal paraffin streams **130** with adsorption based separation.

(19) Having disclosed the basic operation of the process for reverse isomerization of a light naphtha feedstock comprising branched C.sub.4-C.sub.6 paraffins, each step of the embodiments of the process is now provided in further detail.

(20) Light Naphtha Feedstock

(21) The light naphtha feedstock **100** comprises light naphtha. For purposes of this disclosure, light naphtha is considered to generally be paraffinic hydrocarbons containing 5 or 6 carbon atoms. Light naphtha generally has a boiling point range of 90 to 200° F. (approximately 32.2 to 93.3° C.). It is noted that the boiling points of iso-pentane is 28° C., n-pentane is 36° C., iso-hexanes is 50 to 62° C., and n-hexane is 69° C. respectively. However in a refinery, due to inefficient separation, there may be carried over material in a light naphtha stream resulting in a limited amount of higher

carbon number molecules as well. Specifically, depending on the source of light naphtha, cut-points, and reactor efficiency, a light naphtha stream may be composed of normal paraffins and iso-paraffins along with a minority percentage of normal and iso-olefins, saturated and unsaturated naphthenes, and even small portion of aromatic compounds such as benzene and toluene. However, for clarity the light naphtha feedstock **100** as discussed in the present disclosure is explicitly indicated as comprising branched C.sub.5-C.sub.7 paraffins where the C.sub.7 concentration may range from 0.1-5 weight percent (wt. %). In further embodiments, the C.sub.7 concentration in the light naphtha feedstock **100** may be 0.1 to 3 wt. %.

(22) In various embodiments, the light naphtha feedstock **100** may comprise normal paraffins and iso-paraffins from various sources including crude oil, gas condensate, coal liquid, bio fuels, intermediate refinery processes, and their combinations. The intermediate refinery processes may include hydrocracking, hydrotreating, delayed coking, visbreaking, fluid catalytic cracking, and effluent of a residue hydroprocessing unit.

(23) The light naphtha feedstock **100** to the process generally comprises normal and single branched C.sub.4-C.sub.8 paraffins, in certain embodiments normal and single branched C.sub.5-C.sub.7 paraffins, and in further embodiments normal and single branched C.sub.5-C.sub.6 paraffins. The C.sub.4-C.sub.8 paraffins, C.sub.5-C.sub.7 paraffins, or C.sub.5-C.sub.6 paraffins in various embodiments form a significant portion of the light naphtha feedstock **100**. Typically, such feeds have a RON of less than 60. It will be appreciated by one skilled in the art that RON represents the research octane number which is a standardized measurement determined by running a fuel in a test engine with a variable compression ratio under controlled conditions, and comparing the results with those for mixtures of iso-octane and n-heptane.

(24) In one or more embodiments, the light naphtha feedstock **100** consists of C.sub.4 to C.sub.7 hydrocarbons. That is the hydrocarbons provided in the light naphtha feedstock **100** have a carbon number between 4 and 7, inclusive.

(25) In certain embodiments the light naphtha feedstock **100** is a light naphtha mixture having an initial boiling point in the range of about 10° C. to about 30° C. and a final boiling point in the range of about 70° C. to about 110° C. For example, in various embodiments, the light naphtha feedstock **100** has a boiling point range of 10° C. to 100° C., 20° C. to 80° C., or 30° C. to 70° C.

(26) In one or more embodiments, the hydrocarbon feed stream **150** comprising the iso-paraffin stream **120** additionally comprises a second light naphtha feedstock **102**. It will be appreciated that the addition of the second light naphtha feedstock **102** to the hydrocarbon feed stream **150** allows for greater control of the process. Specifically, the ratios of iso-paraffins and normal paraffins entering the reverse isomerization reactor **10** may be adjusted based on the inlet rate of the second light naphtha feedstock **102**. Further, addition of the second light naphtha feedstock **102** allows for flow through the reverse isomerization reactor **10** as measured by the Liquid hourly space velocity (LHSV) to be stabilized regardless of variations in the split between iso-paraffins and normal paraffins and the removal of the one or more normal paraffin streams **130** in the separation unit **30**.

(27) In one or more embodiments, the light naphtha feedstock **100** and the second light naphtha feedstock **102** comprise substantially the same composition. However, it will be appreciated that in one or more embodiments, the light naphtha feedstock **100** and the second light naphtha feedstock **102** may comprise differing and distinct compositions. Regardless, the composition and physical parameters of the second light naphtha feedstock **102** should conform with those defined for the light naphtha feedstock **100**. For example, the second light naphtha feedstock **102** may comprise normal paraffins and single branched C.sub.4-C.sub.8 paraffins, in certain embodiments normal and single branched C.sub.5-C.sub.7 paraffins, and in further embodiments normal and single branched C.sub.5-C.sub.6 paraffins. Further, the second light naphtha feedstock **102** may typically, have a RON of less than 60. Additionally, the second light naphtha feedstock **102** may include light naphtha mixture having an initial boiling point in the range of about 10° C. to about 30° C. and a final boiling point in the range of about 70° C. to about 110° C.

(28) Separation Unit

(29) The process for reverse isomerization of the light naphtha feedstock **100** includes feeding the mixed feed stream **110** to the separation unit **30** to generate the iso-paraffin stream **120** and the one or more normal paraffin streams **130**. The mixed feed stream **110** comprises the light naphtha feedstock **100** and the stabilized isomerate stream **190**. However, it will be appreciated that at initiation of the process for reverse isomerization of the light naphtha feedstock **100** the stabilized isomerate stream **190** has not yet been generated and as such the mixed feed stream **110** may initially consist of only the light naphtha feedstock **100**. Alternatively, if the second light naphtha feedstock **102** is fed to the reverse isomerization reactor **10**, the second light naphtha feedstock **102** may generate an initial supply of the stabilized isomerate stream **190**.

(30) In various embodiments, the separation unit **30** may comprise any separation device that at least partially separates one or more chemicals that are mixed in a process stream from one another. For example, the separation unit **30** may selectively separate differing chemical species from one another, forming one or more chemical fractions. Examples of separation units **30** include, without limitation, distillation columns, flash drums, knock-out drums, knock-out pots, centrifuges, filtration devices, traps, scrubbers, expansion devices, membranes, solvent extraction devices, and the like. It should be understood that separation processes described in this disclosure with regards to the separation unit **30** may not completely separate all of one chemical constituent from all of another chemical constituent. It should be understood that the separation processes described in this disclosure “at least partially” separate different chemical components from one another, and that even if not explicitly stated, it should be understood that separation may include only partial separation. However, even if not explicitly stated, it should be understood that separation may include complete separation in one or more embodiments as well.

(31) It should be additionally understood that where only one separation unit **30** is depicted in a figure or described, two or more separation units **30** may be employed to carry out the identical or substantially identical separation. For example, where a separation unit **30** with multiple outlets is described or illustrated, it is contemplated that several separation units **30** arranged in series may equally separate the feed or feeds to the separation unit **30** and such embodiments are within the scope of the presently described embodiments.

(32) In one or more embodiments and generally with reference to FIGS. **1** and **2**, the separation unit **30** may operate on the basis of thermal separation of the various components of the mixed feed stream **110**. In accordance with such embodiments, the separation unit **30** includes the deisopentanizer column **40** and the fractionator **50**.

(33) In accordance with such embodiments, the mixed feed stream **110** is provided to the deisopentanizer column **40** to separate iso-pentane as the iso-pentane stream **122** from the mixed feed stream **110** which generates a residual stream of the deisopentanizer effluent **200** comprising normal pentane and hexanes. The deisopentanizer effluent **200** is provided to the fractionator **50** to generate the iso-hexanes stream **124**, the normal pentane stream **132**, and the normal hexane stream **134**. It is noted that the iso-pentane stream **122** and the iso-hexanes stream **124** may be combined to generate the iso-paraffin stream **120**.

(34) The deisopentanizer column **40** may include a simple flash column with no theoretical plates or a stripper with a gas or a steam or a single fractionation column with at least 15 theoretical trays or may include a plurality of atmospheric distillation units, vacuum distillation units, or both, which may be operated in series or in parallel to separate the mixed feed stream **110** comprising the light naphtha feedstock **100** and the isomerization effluent stream **190** into the iso-pentane stream **122** and the deisopentanizer effluent **200** comprising normal pentane and hexanes.

(35) It is noted that iso-pentane forming the iso-pentane stream **122** has a boiling point of approximately 28° C. and normal pentane and hexanes forming the deisopentanizer effluent **200** have a boiling point of approximately 36° C. or greater. As such, the split between the iso-pentane stream **122** and the deisopentanizer effluent **200** is made in the range of 28° C. to 36° C. In various

embodiments, the deisopentanizer column **40** may obtain cuts of the mixed feed stream **110** to the deisopentanizer column **40** within $\pm 5^{\circ}\text{C}$., $\pm 4^{\circ}\text{C}$., $\pm 3^{\circ}\text{C}$., or $\pm 2^{\circ}\text{C}$.. Further, the pressure within the deisopentanizer column **40** is typically in the range of 1 to 3 bars.

(36) The fractionator **50** may include a simple flash column with no theoretical plates or a stripper with a gas or a steam or a single fractionation column with at least 15 theoretical trays or may include a plurality of atmospheric distillation units, vacuum distillation units, or both, which may be operated in series or in parallel to separate the deisopentanizer effluent **200** into the iso-hexanes stream **124**, the normal pentane stream **132**, and the normal hexane stream **134**.

(37) It is noted that n-pentane forming the normal pentane stream **132** has a boiling point of approximately 36°C ., iso-hexanes forming the iso-hexanes stream **124** have a boiling point of approximately 50 to 62°C ., and normal hexane forming the normal hexane stream **134** has a boiling point of approximately 69°C . As such, the split between the iso-hexanes stream **124**, the normal pentane stream **132** is made in the range of 36°C . to 50°C . and the split between the iso-hexanes stream **124** and the normal hexane stream **134** is made in the range of 62°C . to 69°C . In various embodiments, the fractionator **50** may obtain cuts of the deisopentanizer effluent **200** provide to the fractionator **50** within $\pm 5^{\circ}\text{C}$., $\pm 4^{\circ}\text{C}$., $\pm 3^{\circ}\text{C}$., or $\pm 2^{\circ}\text{C}$.. Further, the pressure within the fractionator **50** is typically in the range of 1 to 3 bars.

(38) In one or more embodiments and generally with reference to FIGS. **3** and **4**, the separation unit **30** may operate on the basis of adsorption based separation of the various components of the mixed feed stream **110**. In accordance with such embodiments, the separation unit **30** includes a molecular sieve separation column **60**. The molecular sieve separation column **60** separates the mixed feed stream **110** into the iso-paraffin stream **120** and the one or more normal paraffin streams **130** directly with adsorption based separation. Operation of a separation column of similar nature to the molecular sieve separation column **60** is described in U.S. Pat. No. 4,176,053, incorporated herein by reference.

(39) The one or more normal paraffin streams **130** generated by the separation unit **30** comprise primarily normal paraffins. In various embodiments, the one or more normal paraffin streams **130** comprises less than 20 percent by weight of iso-paraffins, less than 15 percent by weight of iso-paraffins, less than 10 percent by weight of iso-paraffins, less than 5 percent by weight of iso-paraffins, or less than 1 percent by weight of iso-paraffins.

(40) The iso-paraffin stream **120** generated by the separation unit **30** comprise primarily iso-paraffins. In various embodiments, the iso-paraffin stream **120** comprises less than 20 percent by weight of normal paraffins, less than 15 percent by weight of normal paraffins, less than 10 percent by weight of normal paraffins, less than 5 percent by weight of normal paraffins, or less than 1 percent by weight of normal paraffins.

(41) Reverse Isomerization Reactor

(42) The process for reverse isomerization of the light naphtha feedstock **100** includes providing the hydrogen-enriched liquid feed stream **160** formed from mixing hydrogen gas **140** and the hydrocarbon feed stream **150** to the reverse isomerization reactor **10**. The reverse isomerization reactor **10** can include one or more fixed-bed, moving-bed, fluidized-bed or batch reactor systems. For example, in one or more embodiments, the reverse isomerization reactor **10** comprises multiple individual reactors in series. In further embodiments, the reverse isomerization reactor **10** comprises an individual reactor. The hydrogen-enriched liquid feed stream **160** provided to the reverse isomerization reactor **10** can be contacted with the solid reverse isomerization catalyst particles **12** in an upward, downward, or radial-flow manner. The reverse isomerization reactor **10** can include a single reactor or multiple reactors with suitable fluid communication between reactors and thermal means and control to ensure that the desired reverse isomerization temperature is maintained at the inlet to each reactor.

(43) In one or more embodiments, the reverse isomerization reactor **10** is operated by contacting the hydrogen-enriched liquid feed stream **160** with the solid reverse isomerization catalyst **12** for

reverse hydroisomerization in a substantially two-phase liquid-solid reverse isomerization fixed-bed reaction zone under conditions that minimize cracking reactions and that are effective to convert iso-paraffins to normal paraffins and recover an equilibrium composition of normal and iso paraffins in the isomerization effluent stream **170**.

(44) In one or more embodiments, conditions within the reverse isomerization reactor **10** are effective to maintain least 90 V % of the hydrogen-enriched liquid feed stream **160** in liquid phase. In various further embodiments, conditions within the reverse isomerization reactor **10** are effective to maintain least 92 V % of the hydrogen-enriched liquid feed stream **160** in liquid phase, maintain least 95 V % of the hydrogen-enriched liquid feed stream **160** in liquid phase, or maintain least 98 V % of the hydrogen-enriched liquid feed stream **160** in liquid phase. It will be appreciated by one skilled in the art that maintaining the hydrogen-enriched liquid feed stream **160** in liquid phase allows the reverse isomerization reactor **10** to operate as a substantially two-phase system with the solid phase of the reverse isomerization catalyst **12** and the liquid phase of the hydrogen-enriched liquid feed stream **160**. It is expressly noted that the hydrogen **140** which is traditionally provided to isomerization or reverse isomerization reactors in a gaseous phase is not present as a gaseous phase in accordance with embodiments of the present disclosure but is instead integrated into the liquid phase. Such arrangement alleviates the problem of gas hold-up that is typical of processed utilizing fixed-bed reactors of traditional design. One skilled in the art will appreciate that gas hold-up represents the reactor volume occupied by gaseous species.

(45) In various embodiments, the reverse isomerization reactor **10** is operated at a temperature of from 20° C. to 300° C., 20° C. to 285° C., 20° C. to 180° C., 50° C. to 300° C., 50° C. to 285° C., 50° C. to 180° C., 80° C. to 300° C., 80° C. to 285° C., 80° C. to 180° C., 100° C. to 300° C., 100° C. to 285° C., or 100° C. to 180° C. Greater reaction temperatures are generally preferred to favor equilibrium mixtures having the highest concentration of normal alkanes. However, the reaction temperature should be constrained to avoid cracking of the hydrogen-enriched liquid feed stream **160** to lighter hydrocarbons at elevated temperatures.

(46) In various embodiments, the reverse isomerization reactor **10** is operated at a pressure of 10 to 100 bars, 10 to 70 bars, 20 to 100 bars, 20 to 70 bars, 30 to 100 bars, or 30 to 70 bars. It will be appreciated that the pressure determines the amount of hydrogen dissolved in the liquid phase with lesser pressures representing less dissolved hydrogen and greater pressure representing more dissolved hydrogen.

(47) In various embodiments, the reverse isomerization reactor **10** is operated at a Liquid hourly space velocity (LHSV) range from about 0.2 to 20 h.sup.-1, 0.2 to 2 h.sup.-1, 1 to 20 h.sup.-1, or 1 to 2 h.sup.-1.

(48) In one or more specific embodiments, the reverse isomerization reactor **10** is operated at conditions effective to maintain at least 98 V % of the hydrogen-enriched liquid feed stream **160** in liquid phase and include a temperature of from 100° C. to 180° C., a pressure of from 20 bars to 70 bars, and a liquid hourly space velocity of 0.2 to 20 h.sup.-1, with a hydrogen to hydrocarbon mole ratio of 0.01 to 20.0 in the hydrogen-enriched liquid feed stream **160**.

(49) Effective catalysts for use in the reverse isomerization reactor **10** include those known to persons having ordinary skill in the art. Reverse isomerization catalysts include, but are not limited to, those that are zeolitic or amorphous such as catalysts based upon amorphous alumina. Example solid reverse isomerization catalysts **12** may include platinum on alumina, zeolite, chlorinated alumina, a sulfated zirconia and platinum, a platinum group metal on chlorided alumina, or a sulfated zirconia. In one or more embodiments the solid reverse isomerization catalyst **12** comprises 0.05 wt. % to 5 wt. % of at least one Group VIIIB metal. In one or more further embodiments the solid reverse isomerization catalyst **12** comprises a base material including zeolite and metal oxides with metals from Group IIIA-B or IVA-B. It will be appreciated that the operating parameters of the reverse isomerization reactor **10** including temperature, pressure, and LHSV may depend upon the specific solid reverse isomerization catalyst **12** utilized within the reverse

isomerization reactor **10**. Conversely, the solid reverse isomerization catalyst **12** may be selected to be compatible with the standard operating parameters of the reverse isomerization reactor **10**.

(50) It will be appreciated that in one or more embodiments, operation of the reverse isomerization reactor **10** in accordance with the present disclosure may be effective to increase the RON for a typical feedstock of the light naphtha feedstock **100** to a RON of at least 80. Specifically, the typical feedstock of the light naphtha feedstock **100** has a RON of 60 or lower and treatment of the light naphtha feedstock **100** with the reverse isomerization reactor **10** increased the RON to at least 80, in certain embodiments at least 85, and in further embodiments at least 90.

(51) In one or more embodiments, the light naphtha feedstock **100** is dried before providing to the separation unit **30**. It will be appreciated that it is desirable to dry the light naphtha feedstock **100** as the potential exists for acid formation leading to catalyst poisoning and metal corrosion when residual moisture is retained. Additionally, in one or more embodiments, the iso-paraffin stream **120** is dried before providing to the reverse isomerization reactor **10**. Similarly, in one or more embodiments, the second light naphtha feedstock **102** is dried before providing to the reverse isomerization reactor **10**. Further, in one or more embodiments, the hydrogen gas **140** is dried before providing to the reverse isomerization reactor **10**. It will be appreciated that the light naphtha feedstock **100**, the second light naphtha feedstock **102**, the iso-paraffin stream **120**, and the hydrogen gas **140** may be dried separately or as the combined hydrogen-enriched liquid feed stream **160** before passage to the reverse isomerization reactor **10**.

(52) In various embodiments, the hydrogen-enriched liquid feed stream **160** comprises less than 0.1 parts per million by weight (ppmw) of water, less than 0.05 ppmw of water, less than 0.03 ppmw of water, less than 0.02 ppmw of water, less than 0.01 ppmw of water, less than 0.001 ppmw of water, or less than 0.001 ppmw of water. It will be appreciated that the hydrogen-enriched liquid feed stream **160** is typically substantially free of water in accordance with the present disclosure.

(53) The sulfur and nitrogen content of the hydrogen-enriched liquid feed stream **160** may be reduced from the levels in the light naphtha feedstock **100** and the second light naphtha feedstock **102**. Reduction of sulfur and nitrogen is desirable to prevent or alleviate deactivation of the solid reverse isomerization catalyst **12** or other catalysts in downstream refinery processes. In one or more embodiments, the hydrogen-enriched liquid feed stream **160** comprises less than 0.5 ppmw sulfur and less than 0.5 ppmw nitrogen when provided to the reverse isomerization reactor **10**. In various embodiments, the hydrogen-enriched liquid feed stream **160** comprises less than 0.4 ppmw sulfur, less than 0.3 ppmw sulfur, less than 0.2 ppmw sulfur, or less than 0.1 ppmw sulfur. In various embodiments, the hydrogen-enriched liquid feed stream **160** comprises less than 0.4 ppmw nitrogen, less than 0.3 ppmw nitrogen, less than 0.2 ppmw nitrogen, or less than 0.1 ppmw nitrogen. It will be appreciated that the disclosed embodiments for the sulfur present in the hydrogen-enriched liquid feed stream **160** and the disclosed embodiments for the nitrogen present in the hydrogen-enriched liquid feed stream **160** are not necessarily exclusive and may be combined in any combination in various embodiments.

(54) In one or more embodiments, the light naphtha feedstock **100** is processed to reduce the sulfur and nitrogen content before the light naphtha feedstock **100** is provided to the separator **30**. Similarly, in one or more embodiments, the second light naphtha feedstock **102** is additionally or alternatively hydrotreated to reduce the sulfur and nitrogen content before the second light naphtha feedstock **102** is provided to the reverse isomerization reactor **10**. It will be appreciated that removal of sulfur and nitrogen from both the light naphtha feedstock **100** and the second light naphtha feedstock **102** may be concurrently achieved by processing the hydrocarbon feed stream **150** before mixing the hydrogen gas **140** with the hydrocarbon feed stream **150** to form the hydrogen-enriched liquid feed stream **160**. Sulfur removal is achieved in a hydrodesulfurization process performed in a "HDS unit" or "hydrotreater" (not shown). Generally, the hydrodesulfurization reaction takes place in a fixed-bed reactor at elevated temperatures of 200-425° C., sometimes more precisely 300-400° C., and elevated pressures of 1-20 MPa gauge, sometimes more precisely 1-13 MPa gauge, in the

presence of a catalyst in a sulfide form comprising elements selected from the group consisting of Ni, Mo, Co, W and Pt, with or without promoters, supported on alumina. Similarly, nitrogen is removed through hydrodenitrogenation reactions in the fixed-bed reactor.

(55) Stabilization Column

(56) The process for reverse isomerization of the light naphtha feedstock **100** includes providing the isomerization effluent stream **170** to the stabilization column **20** to separate the isomerization effluent stream **170** into the C.sub.1-C.sub.4 hydrocarbon stream **180** and the stabilized isomerate stream **190**. The stabilization column **20** removes or separates out the lighter components in the isomerization effluent stream **170** generated in the reverse isomerization reactor **10** as the C.sub.1-C.sub.4 hydrocarbon stream **180** to generate the stabilized isomerate stream **190**. The stabilized isomerate stream **190** thus has a higher concentration of desirable C.sub.5-C.sub.6 hydrocarbons for collection as normal paraffins.

(57) As the general operation of stabilization columns is known to one skilled in the art the reboilers, condensers, reflux lines, and other typical components of a stabilization column are not expressly illustrated in the Figures. However, it will be appreciated that, as appropriate, such components are contemplated as forming part of the stabilization column **20**.

(58) In one or more embodiments, the stabilization column **20** is operated at a pressure of 1 to 3 bar and a temperature of 120 to 140° C.

(59) Hydrogen and Hydrocarbon Mixing Zone

(60) The process for reverse isomerization of the light naphtha feedstock **100** includes mixing the hydrogen gas **140** and the hydrocarbon feed stream **150** comprising the iso-paraffin stream **120** to form the hydrogen-enriched liquid feed stream **160**. With reference to all the Figures and specific reference to FIGS. 5 through 8, mixing the hydrogen gas **140** and the hydrocarbon feed stream **150** is completed in a mixing zone **70**. The mixing zone **70** has at least one inlet for receiving the hydrocarbon feed stream **150** and at least one inlet for receiving the hydrogen gas **140** and an outlet for discharging the hydrogen-enriched liquid feed stream **160**. It will be appreciated that the at least one inlet for receiving the hydrocarbon feed stream **150** and at least one inlet for receiving the hydrogen gas **140** may be combined into a single combined inlet for receiving both the hydrocarbon feed stream **150** the hydrogen gas **140**.

(61) In one or more embodiments, the mixing zone **70** may simply be the piping carrying the hydrocarbon feed stream **150** such that the hydrogen gas **140** is added directly to the hydrocarbon feed stream **150** and integrated through turbulence in the flow before entering the isomerization reactor **10**. Hydrogen can be added at the solubility limit to prevent any gas phase hydrogen rendering a simpler solution for a low hydrogen consumption system.

(62) During operation the system and execution of the process for reverse isomerization of the light naphtha feedstock **100**, the hydrocarbon feed stream **150** is intimately mixed with the hydrogen gas **140** in the mixing zone **70** to dissolve a predetermined quantity of the hydrogen gas **140** in the light naphtha feedstock **100** and produce the hydrogen-enriched liquid feed stream **160**. The hydrogen gas **140** includes fresh hydrogen introduced via stream **142** and optionally recycled hydrogen introduced via stream **144** from an optional flashing zone **76** which removes excess hydrogen. In addition, in certain embodiments hydrogen can be recycled after separation from the isomerization effluent stream **170** in the stabilization column **20**.

(63) In one or more embodiments in which excess hydrogen gas **140** is combined with the hydrocarbon feed stream **150** in the mixing zone **62** and with reference to FIGS. 6 and 8, the hydrogen-enriched liquid feed stream **160** is conveyed to a flashing zone **74** in which undissolved hydrogen and other gases such as light feedstock fractions may be flashed off and removed as stream **162**. The heavier fraction **164**, which is the hydrogen-enriched liquid feed stream **160** with excess hydrogen removed, serves as the feed to the reverse isomerization reactor **10**.

(64) A portion **144** of stream **162** can optionally be recycled and mixed with the fresh hydrogen feed **142**. The amount of recycled hydrogen in the hydrogen gas **140** generally depends upon a

variety of factors relating to the excess undissolved hydrogen recovered from the flashing zone **74**. For instance, the amount of stream **144** relative to stream **162** can be in the range of about 50 to 100 V %, 50 to 99.5 V %, 50 to 99 V %, 50 to 95 V %, 80 to 100 V %, 80 to 99.5 V %, 80 to 99 V %, 80 to 95 V %, 90 to 100 V %, 90 to 99.5 V %, 90 to 99 V %, or 90 to 95 V %. A remaining portion of the flashed gases can be discharged from the system as a bleed stream **164**, in embodiments where portion **144** is not 100 V % of stream **162**. Bleed stream **164**, the portion of stream **162** not recycled as stream **144**, is effective to remove accumulated impurities.

(65) The mixing zone **70** can be any apparatus that achieves the necessary intimate mixing of liquid and gas so that a sufficient amount of the hydrogen gas **140** is dissolved in the hydrocarbon feed stream **150**. The mixing zone **70** can include a combined inlet for the hydrogen gas **140** and the hydrocarbon feed stream **150** or separate inlets as depicted in FIGS. 5 through 8. Mixing zone **70** can be a separate vessel or unit operation, in-line mixing apparatus, or a combination thereof to achieve the requisite hydrogen saturation.

(66) Effective unit operations for a hydrogen distributor vessel **72** in the mixing zone **70** include one or more gas-liquid distributor vessels, which apparatus can include spargers, injection nozzles, or other devices that impart sufficient velocity to inject the hydrogen gas **140** into the hydrocarbon feed stream **150** with turbulent mixing and thereby promote hydrogen saturation. Suitable apparatus are described with respect to FIGS. 5 through 8 herein, and also, for instance, in U.S. Pat. Nos. 3,378,349; 3,598,541; 3,880,961; 4,960,571; 5,158,714; 5,484,578; 5,837,208; 5,942,197; and 11,155,757, the relevant portions of which are incorporated herein by reference.

(67) In certain embodiments, such as, for example, shown in FIGS. 7 and 8, a column is used as the hydrogen distributor vessel **72**, in which the hydrogen gas **140** is injected at plural locations **140a**, **140b**, **140c**, **140d** and **140e**. The hydrogen gas **140** is injected through hydrogen distributors into the column for adequate mixing to effectively dissolve the hydrogen gas **140** in the hydrocarbon feed stream **150**. For instance, suitable injection nozzles can be provided proximate several plates (locations **112a-112d**) and also at the bottom of the column (location **112e**). The hydrocarbon feed stream **150** can be fed from the top of the column as shown in the figures or from the bottom of the column (not shown).

(68) Various types of hydrogen distributor apparatus can be used. For instance, the gas distributor apparatuses can include one or both of tubular injectors fitted with nozzles and jets that are configured to uniformly distribute hydrogen gas into the flowing hydrocarbon feed stream **150** in a column or vessel in order to achieve a saturation state in the hydrogen distributor vessel **72**.

(69) Operating conditions in the mixing zone **70** and more particularly the hydrogen distributor vessel **72** are selected to promote solubility of the hydrogen gas **140** within the hydrocarbon feed stream **150**. The mixing zone **70** is maintained at pressure levels of from about 5 bars to about 200 bars, and at a ratio of the normalized volume of hydrogen to the volume of liquid hydrocarbon of about 30 to about 300 normalized liters of hydrogen per liter of liquid hydrocarbon.

(70) The optional flashing zone **74** can include one or more flash drums that are maintained at suitable operating conditions to maintain an effective amount of the hydrogen gas **140** in solution in the hydrogen-enriched liquid feed stream **160**.

(71) In various embodiments, the generated hydrogen-enriched liquid feed stream **160** is sent to the reverse isomerization reactor **10** at a hydrogen to hydrocarbon mole ratio of from about 0.01 to 20, 0.01 to 10, 0.01 to 1, 0.02 to 20, 0.02 to 10, 0.02 to 1, 0.05 to 20, 0.05 to 10, or 0.05 to 1. In one or more embodiments, the hydrogen-enriched liquid feed stream **160** is saturated with hydrogen.

(72) Saturation Reactor

(73) In one or more embodiments, the hydrogen-enriched liquid feed stream **160** is provided to a saturation reactor **80** upstream of the reverse isomerization reactor **10** to convert benzene in the hydrogen-enriched liquid feed stream **160** to cyclohexene prior to introduction to the reverse isomerization reactor **10**. In one or more embodiments, the hydrocarbon feed stream **150** is provided to the saturation reactor **80** upstream of the mixing zone **70** to convert benzene in the

hydrocarbon feed stream **150** to cyclohexene prior to introduction to the reverse isomerization reactor **10**. One skilled in the art will appreciate that benzene in the presence of hydrogen over a platinum-containing catalyst will generate heat and saturate to cyclohexene. Such benzene saturation reaction is fast and happens near the inlet of the reverse isomerization reactor **10** if no saturation reactor **80** is provided to pretreat the hydrogen-enriched liquid feed stream **160** before the reverse isomerization reactor **10**. As such, the saturation reactor **80** may be placed upstream of the reverse isomerization reactor **10** to convert benzene in the hydrogen-enriched liquid feed stream **160** to cyclohexene prior introduction to the reverse isomerization reactor **10**. Further, in one or more embodiments, the saturation reactor **80** may be located upstream of the mixing zone **70** such that benzene in hydrocarbon feed stream **150** is converted to cyclohexene prior to passage to the mixing zone **70** and introduction of the hydrogen gas **140**.

(74) It will be appreciated that it is desirable convert benzene to cyclohexene prior to introduction to the isomerization reactor **10** as cyclic compounds may be adsorbed on the catalyst in the isomerization reactor **10** which reduces the active sites available for paraffin isomerization. As such, if the feed to the isomerization reactor **10** contains significant amounts of cyclic compounds, such as benzene, the catalyst inventory in the isomerization reactor **10** has to be increased.

Unsaturated cyclic hydrocarbons consume considerable quantities of hydrogen, resulting in exothermic reactions disrupting preferred isomerization equilibrium. Without wishing to be bound by theory, it is believed that each 1 vol. % of benzene leads to 10° C. exotherm. As such, with high benzene content, the isomerization reaction temperature may be undesirably increased.

(75) In one or more embodiments, the one or more normal paraffin streams **130** are provided to a steam cracking unit for ethylene production. In further embodiments, the one or more normal paraffin streams **130**, the C.sub.1-C.sub.4 hydrocarbon stream **180**, or both are provided to further unit operations for further processing into useful or value added compositions.

EXAMPLES

(76) The following examples illustrate features of the present disclosure but are not intended to limit the scope of the disclosure.

(77) To verify and demonstrate the normal paraffin recovery with processes and systems in accordance with the present disclosure, a material balance model was prepared for a processing scheme in accordance with the present disclosure. Specifically, a light straight run naphtha stream in accordance with the light naphtha feedstock **100** was processed. Specifically, light straight run naphtha composed of pentanes and hexanes with iso to normal weight ratio of 42.9:52.7 with contaminants of less than 0.5 ppmw of sulfur and less than 0.1 ppmw of nitrogen and oxygenates was used as a feed. It is noted that water present in the feed was separated from the feed during subsequent fractionation of the feed and the feed was dried before being provided to the isomerization reactor. The composition of the light straight run naphtha stream was in accordance with the composition of Table 1.

(78) TABLE-US-00001

TABLE 1 Light Naphtha Feedstock Composition (kg/hr)		Component	Light Naphtha Feedstock
Hydrogen	0.0	Water	0.1
i-butan	0.0	n-butan	1.1
i-pentan	19.0	n-pentan	32.8
2,2-dimethylbutan	0.2	2,3-dimethylbutan	3.4
2-methylpentan	12.5	3-methylpentan	7.8
n-hexan	18.8	methycyclopentan	2.2
cyclohexan	0.6	benzen	1.4
n-heptan	0.2	TOTAL	100.0

(79) The light straight run naphtha stream representing the light naphtha feedstock **100** and a stream representing the stabilized isomerate stream **190** were combined to form a stream representing the mixed feed stream **110**. The composition of the stabilized isomerate stream **190** and the mixed feed stream **110** are detailed in Table 2.

(80) TABLE-US-00002

TABLE 2 Stabilized Isomerate Stream and Mixed Feed Stream Compositions (kg/hr)		Stabilized Isomerate Stream	Mixed Feed Stream
Hydrogen	0.0	0.0	0.0
Water	0.0	0.1	0.1
i-butan	0.3	0.3	0.6
n-butan	1.6	1.6	1.6
i-pentan	25.4	44.4	8.0
n-pentan	8.0	40.8	2.2
2,2-dimethylbutan	8.6	8.8	2.8
2,3-dimethylbutan	2.8	6.2	2.8
2-methylpentan	8.8	21.3	4.6
3-methylpentan	4.6	12.4	2.8
n-hexan	2.8	21.7	2.9
methycyclopentan	0.8	0.8	1.4
cyclohexan	0.8	1.4	0.0
benzen	0.0	1.4	0.0
n-heptan	0.0	0.0	0.0

0.4 0.5 TOTAL 63.8 163.8

(81) The mixed feed stream **110** was provide to a deisopentantanizer **40** in accordance with the present disclosure to generate the iso-pentane stream **122** and the deisopentanizer effluent **200**. The composition of the iso-pentane stream **122** and the deisopentanizer effluent **200** are detailed in Table 3.

(82) TABLE-US-00003 TABLE 3 Iso-Pentane Stream and Deisopentanizer Effluent Compositions (kg/hr)

	Iso-Pentane	Deisopentanizer	Component	Stream	Effluent	Hydrogen	0.0	0.0	Water	0.1	0.0
i-butan	0.3	0.0	n-butan	1.6	0.0	i-pentan	42.2	2.2	n-pentan	2.0	38.7
2.2			dimethy	butan	0.4	8.4	2.3	dimethy	butan	0.3	5.9
2.0			2.methy	pentan	1.1	20.2	3.methy	pentan	0.6	11.8	
2.2			n-hexan	1.1	20.6	methycy	clopent	0.1	2.8	cyclohex	0.1
2.2			benzen	0.1	1.3	n-heptan	0.0	0.5	TOTAL	50.1	113.7

(83) The deisopentanizer effluent **200** was provide to a fractionator **50** in accordance with the present disclosure and operating at 100° C. and 3 bars to generate the iso-hexanes stream **124**, the normal pentane stream **132**, and the normal hexane stream **134**. The composition of the iso-hexanes stream **124**, the normal pentane stream **132**, and the normal hexane stream **134** are detailed in Table 4.

(84) TABLE-US-00004 TABLE 4 Iso-Hexanes Stream and Normal Paraffin Streams Compositions (kg/hr)

	Iso-	Normal	Normal	Hexanes	Pentane	Hexane	Component	Stream	Stream	Stream
Hydrogen	0.0	0.0	0.0	Water	0.0	0.0	0.0	i-butan	0.0	0.0
0.0				0.0	0.0	0.0	0.0	n-butan	0.0	0.0
0.0				0.0	0.0	0.0	0.0	i-pentan	2.2	0.0
0.0				1.9	36.8	0.0	2.2	dimethy	butan	7.9
0.4				0.0	2.3	dimethy	butan	5.6	0.3	0.0
2.0				2.methy	pentan	19.2	1.0	0.0	3.methy	pentan
11.2				0.6	0.0	n-hexan	0.0	1.0	19.6	methycy
0.0				0.0	0.1	2.7	cyclohex	0.0	0.1	1.2
0.0				0.0	0.1	1.2	benzen	0.0	0.1	1.2
0.0				0.0	0.0	0.5	TOTAL	48.1	40.4	25.2

(85) The iso-pentane stream **122** and the iso-hexanes stream **124** were combined with hydrogen gas at a hydrogen to hydrocarbon mole ratio of 0.05 to generate the hydrogen-enriched liquid feed stream **160** fed to the isomerization reactor **10**. The composition of the hydrogen-enriched liquid feed stream **160** is detailed in Table 5.

(86) TABLE-US-00005 TABLE 5 Hydrogen and Hydrogen-Enriched Liquid Feed Stream Compositions (kg/hr)

	Hydrogen	Hydrogen-Enriched	Component	Stream	Liquid	Feed	Stream
Hydrogen	1.5	1.5	Water	0.0	0.1	i-butan	0.0
0.0			0.3	n-butan	0.0	1.6	i-pentan
0.0			0.0	44.4	n-pentan	0.0	4.0
2.2			2.2	dimethy	butan	0.0	8.4
2.3			2.3	dimethy	butan	0.0	5.9
2.0			2.0	3.methy	pentan	0.0	20.3
11.8			11.8	n-hexan	0.0	1.1	methycy
0.0			0.0	0.1	cyclohex	0.0	0.1
0.0			0.0	0.1	benzen	0.0	0.1
0.0			0.0	0.0	TOTAL	1.5	99.7

(87) The hydrogen-enriched liquid feed stream **160** was fed to the isomerization reactor **10** and processed according to the present disclosure. Specifically, the isomerization reactor **10** was operated with a platinum on chlorinated alumina catalyst at a temperature of 100° C., pressure of 50 bars, and LHSV of 1.6 h.sup.-1. After further processing in a stabilization column **20**, the stabilized isomerate stream **190** as detailed in Table 2 was generated.

(88) It will be appreciated that 65.6 kg/h of normal hexanes were generated (combination of the normal pentane stream **132** and the normal hexane stream **134** as detailed in Table 4) from the input of 100 kg/h of the light naphtha feedstock. As the light naphtha feedstock had an iso to normal weight ratio of 50:50 this represents a significant shift toward normal paraffins.

(89) It should now be understood the various aspects of the process and associate system for reverse isomerization of a light naphtha feedstock comprising branched C.sub.5-C.sub.7 paraffins are described and such aspects may be utilized in conjunction with various other aspects.

(90) According to a first aspect, a process for reverse isomerization of a light naphtha feedstock comprising branched C.sub.5-C.sub.7 paraffins includes feeding a mixed feed stream comprising the light naphtha feedstock to a separation unit to generate an iso-paraffin stream and one or more normal paraffin streams; mixing hydrogen gas and a hydrocarbon feed stream comprising the iso-paraffin stream to form a hydrogen-enriched liquid feed stream; providing the hydrogen-enriched

liquid feed stream to a reverse isomerization reactor; operating the reverse isomerization reactor by contacting the hydrogen-enriched liquid feed stream with a solid reverse isomerization catalyst for reverse hydroisomerization in a substantially two-phase liquid-solid reverse isomerization fixed-bed reaction zone under conditions that minimize cracking reactions and that are effective to convert iso-paraffins to normal paraffins and recovering an equilibrium composition of normal and iso paraffins in an isomerization effluent stream; providing the isomerization effluent stream to a stabilization column to separate the isomerization effluent stream into a C.sub.1-C.sub.4 hydrocarbon stream and a stabilized isomerate stream; and combining the stabilized isomerate stream with the light naphtha feedstock to generate the mixed feed stream.

(91) A second aspect includes the process of the first aspect in which the hydrocarbon feed stream additionally comprises a second light naphtha feedstock.

(92) A third aspect includes the process of the second aspect the light naphtha feedstock and the second light naphtha feedstock comprise substantially the same composition.

(93) A fourth aspect includes the process of any of the first through third aspects in which the light naphtha feedstock consists of C.sub.4 to C.sub.7 hydrocarbons.

(94) A fifth aspect includes the process of any of the first through fourth aspects in which the light naphtha feedstock has a boiling range of 10 to 100° C.

(95) A sixth aspect includes the process of any of the first through fourth aspects in which the light naphtha feedstock has a boiling range of 20 to 80° C.

(96) A seventh aspect includes the process of any of the first through sixth aspects in which the light naphtha feedstock has a RON of 60 or less.

(97) An eighth aspect includes the process of any of the first through seventh aspects in which the isomerate effluent stream has a RON of at least 80.

(98) A ninth aspect includes the process of any of the first through eighth aspects in which the light naphtha feedstock is dried before providing to the separation unit.

(99) A tenth aspect includes the process of any of the first through ninth aspects in which the iso-paraffin stream is dried before providing to the reverse isomerization reactor.

(100) An eleventh aspect includes the process of any of the first through tenth aspects in which the second light naphtha feedstock is dried before providing to the reverse isomerization reactor.

(101) A twelfth aspect includes the process of any of the first through eleventh aspects in which the hydrogen gas is dried before providing to the reverse isomerization reactor.

(102) A thirteenth aspect includes the process of any of the first through twelfth aspects in which the hydrogen-enriched liquid feed stream comprises less than 0.05 ppmw of water.

(103) A fourteenth aspect includes the process of any of the first through thirteenth aspects in which the hydrogen-enriched liquid feed stream comprises less than 0.5 ppmw sulfur and less than 0.5 ppmw nitrogen.

(104) A fifteenth aspect includes the process of the fourteenth aspect in which the light naphtha feedstock is hydrotreated to reduce the sulfur and nitrogen content.

(105) A sixteenth aspect includes the process of the fourteenth or fifteenth aspect s in which the second light naphtha feedstock, if present, is hydrotreated to reduce the sulfur and nitrogen content.

(106) A seventeenth aspect includes the process of any of the first through sixteenth aspects in which the reverse isomerization reactor is operated at a temperature of from 20° C. to 300° C.

(107) An eighteenth aspect includes the process of any of the first through seventeenth aspects in which the reverse isomerization reactor is operated at a temperature of from 100° C. to 180° C.

(108) A nineteenth aspect includes the process of any of the first through eighteenth aspects in which the reverse isomerization reactor is operated at a pressure of from 10 bars to 100 bars.

(109) A twentieth aspect includes the process of any of the first through nineteenth aspects in which the reverse isomerization reactor is operated at a pressure of from 20 bars to 70 bars.

(110) A twenty-first aspect includes the process of any of the first through twentieth aspects in which the reverse isomerization reactor is operated at a LHSV of 0.2 to 20 h.sup.-1.

- (111) A twenty-second aspect includes the process of any of the first through twenty-first aspects in which the reverse isomerization reactor is operated at a LHSV of 1 to 2 h.sup.-1.
- (112) A twenty-third aspect includes the process of any of the first through twenty-second aspects in which the hydrogen-enriched liquid feed stream comprises a hydrogen to hydrocarbon mole ratio of 0.01 to 20.0.
- (113) A twenty-fourth aspect includes the process of any of the first through twenty-third aspects in which the hydrogen-enriched liquid feed stream comprises a hydrogen to hydrocarbon mole ratio of 0.02 to 10.0.
- (114) A twenty-fifth aspect includes the process of any of the first through twenty-fourth aspects in which the hydrogen-enriched liquid feed stream is saturated with hydrogen.
- (115) A twenty-sixth aspect includes the process of any of the first through twenty-fifth aspects in which the conditions within the reverse isomerization reactor are effective to maintain least 90 V % of the hydrogen-enriched liquid feed stream in liquid phase.
- (116) A twenty-seventh aspect includes the process of any of the first through twenty-sixth aspects in which the conditions within the reverse isomerization reactor are effective to maintain least 95 V % of the hydrogen-enriched liquid feed stream in liquid phase.
- (117) A twenty-eighth aspect includes the process of any of the first through twenty-seventh aspects in which the conditions within the reverse isomerization reactor are effective to maintain least 98 V % of the hydrogen-enriched liquid feed stream in liquid phase.
- (118) A twenty-ninth aspect includes the process of any of the first through twenty-eighth aspects in which the solid reverse isomerization catalyst comprises 0.05 wt. % to 5 wt. % of at least one Group VIIIB metal.
- (119) A thirtieth aspect includes the process of any of the first through twenty-ninth aspects in which the solid reverse isomerization catalyst comprises a base material including zeolite and metal oxides with metals from Group IIIA-B or IVA-B.
- (120) A thirty-first aspect includes the process of any of the first through thirtieth aspects in which the solid reverse isomerization catalyst is chlorinated.
- (121) A thirty-second aspect includes the process of any of the first through thirty-first aspects in which the reverse isomerization reactor comprises multiple individual reactors in series.
- (122) A thirty-third aspect includes the process of any of the first through thirty-second aspects in which mixing the hydrocarbon feed stream and the hydrogen gas to form the hydrogen-enriched liquid feed stream is achieved in a mixing zone that comprises one or more gas-liquid distributor vessels that include a plurality of hydrogen distribution apparatus, each hydrogen distribution apparatus comprising a tubular injector fitted with a nozzle, a jet, or a nozzle and jet and that is configured to uniformly distribute hydrogen gas into the hydrocarbon feed stream to achieve a saturation state in the mixing zone.
- (123) A thirty-fourth aspect includes the process of the thirty-third aspect in which the one or more gas-liquid distributor vessels is a column having a top, a bottom and plural plates, and wherein a hydrogen distribution apparatus is included at the bottom and at each of the plates.
- (124) A thirty-fifth aspect includes the process of any of the first through thirty-fourth aspects in which the reverse isomerization reactor is operated at conditions effective to maintain at least 98 V % of the hydrogen-enriched liquid feed stream in liquid phase and include a temperature of from 100° C. to 180° C., a pressure of from 20 bars to 70 bars, and a liquid hourly space velocity of 0.2 to 20 h.sup.-1, with a hydrogen to hydrocarbon mole ratio of 0.01 to 20.0 in the hydrogen-enriched liquid feed stream.
- (125) A thirty-sixth aspect includes the process of any of the first through thirty-fifth aspects in which the hydrogen-enriched liquid feed stream is provided to a saturation reactor upstream of the reverse isomerization reactor to convert benzene in the hydrogen-enriched liquid feed stream to cyclohexene prior introduction to the reverse isomerization reactor.
- (126) A thirty-seventh aspect includes the process of any of the first through thirty-sixth aspects in

which the one or more normal paraffin streams comprises less than 20 percent by weight of iso-paraffins.

(127) A thirty-eighth aspect includes the process of any of the first through thirty-seventh aspects in which the one or more normal paraffin streams comprises less than 5 percent by weight of iso-paraffins.

(128) A thirty-ninth aspect includes the process of any of the first through thirty-eighth aspects in which the one or more normal paraffin streams comprises less than 1 percent by weight of iso-paraffins.

(129) A fortieth aspect includes the process of any of the first through thirty-ninth aspects in which the separation unit comprises a deisopentanizer column and a fractionator, wherein: the mixed feed stream is provided to the deisopentanizer column to separate iso-pentane as an iso-pentane stream from the mixed feed stream to generate a deisopentanizer effluent comprising normal pentane and hexanes, the deisopentanizer effluent is provided to the fractionator to generate an iso-hexanes stream, a normal pentane stream, and a normal hexane stream; and combining the iso-pentane stream and the iso-hexanes stream to generate the iso-paraffin stream.

(130) A forty-first aspect includes the process of any of the first through thirty-ninth aspects in which the separation unit comprises a molecular sieve separation column, wherein the molecular sieve separation column separates the mixed feed stream into the iso-paraffin stream and the one or more normal paraffin streams with adsorption based separation.

(131) A forty-second aspect includes the process of any of the first through forty-first aspects in which the one or more normal paraffin streams are provided to a steam cracking unit for ethylene production.

(132) A forty-third includes the process of any of the first through forty-second aspects in which the light naphtha feedstock is limited to 0.1 to 5 wt. % of C.sub.7 hydrocarbons.

(133) A forty-four includes the process of any of the first through forty-third aspects in which the light naphtha feedstock is limited to 0.1 to 3 wt. % of C.sub.7 hydrocarbons.

(134) It should be apparent to those skilled in the art that various modifications and variations can be made to the described embodiments without departing from the spirit and scope of the claimed subject matter. Thus, it is intended that the specification cover the modifications and variations of the various described embodiments provided such modifications and variations come within the scope of the appended claims and their equivalents.

(135) The verb “comprises” and its conjugated forms should be interpreted as referring to elements, components or steps in a non-exclusive manner. The referenced elements, components or steps may be present, utilized or combined with other elements, components or steps not expressly referenced. For the purposes of defining the present technology, the transitional phrase “consisting of” may be introduced in the claims as a closed preamble term limiting the scope of the claims to the recited components or steps and any naturally occurring impurities. For the purposes of defining the present technology, the transitional phrase “consisting essentially of” may be introduced in the claims to limit the scope of one or more claims to the recited elements, components, materials, or method steps as well as any non-recited elements, components, materials, or method steps that do not materially affect the novel characteristics of the claimed subject matter. The transitional phrases “consisting of” and “consisting essentially of” may be interpreted to be subsets of the open-ended transitional phrases, such as “comprising” and “including,” such that any use of an open ended phrase to introduce a recitation of a series of elements, components, materials, or steps should be interpreted to also disclose recitation of the series of elements, components, materials, or steps using the closed terms “consisting of” and “consisting essentially of.” For example, the recitation of a composition “comprising” components A, B, and C should be interpreted as also disclosing a composition “consisting of” components A, B, and C as well as a composition “consisting essentially of” components A, B, and C. Any quantitative value expressed in the present application may be considered to include open-ended embodiments consistent with the transitional phrases

“comprising” or “including” as well as closed or partially closed embodiments consistent with the transitional phrases “consisting of” and “consisting essentially of.”

(136) It is noted that one or more of the following claims utilize the term “where” as a transitional phrase. For the purposes of defining the present technology, it is noted that this term is introduced in the claims as an open-ended transitional phrase that is used to introduce a recitation of a series of characteristics of the structure and should be interpreted in like manner as the more commonly used open-ended preamble term “comprising.”

(137) It should be understood that any two quantitative values assigned to a property may constitute a range of that property, and all combinations of ranges formed from all stated quantitative values of a given property are contemplated in this disclosure. That is, it is envisioned that each discrete value encompassed by the ranges are also included. Additionally, the ranges which may be formed by each discrete value encompassed by the explicitly disclosed ranges are equally envisioned. For brevity, the same is not explicitly indicated subsequent to each disclosed range and the present general indication is provided.

(138) Throughout the disclosure there is indication of specific parameters being “less than” a value without a specified lower bound. It will be appreciated that such indication implicitly includes the range from zero to the specified value.

(139) Having described the subject matter of the present disclosure in detail and by reference to specific embodiments, it is noted that the various details described in this disclosure should not be taken to imply that these details relate to elements that are essential components of the various embodiments described in this disclosure, even in cases where a particular element is illustrated in each of the drawings that accompany the present description. Rather, the claims appended hereto should be taken as the sole representation of the breadth of the present disclosure and the corresponding scope of the various embodiments described in this disclosure. Further, it will be apparent that modifications and variations are possible without departing from the scope of the appended claims.

Claims

1. A process for reverse isomerization of a light naphtha feedstock comprising branched C.sub.5-C.sub.7 paraffins, the process comprising: feeding a mixed feed stream comprising the light naphtha feedstock to a separation unit to generate an iso-paraffin stream and one or more normal paraffin streams; mixing hydrogen gas and a hydrocarbon feed stream comprising the iso-paraffin stream in a mixing zone to fully dissolve the hydrogen gas in the hydrocarbon feed stream to form a hydrogen-enriched liquid feed stream; providing the hydrogen-enriched liquid feed stream to one or more reverse isomerization reactors, wherein the hydrogen-enriched liquid feed stream is provided to a saturation reactor upstream of the reverse isomerization reactor to convert benzene in the hydrogen-enriched liquid feed stream to cyclohexene prior introduction to the reverse isomerization reactor; operating the reverse isomerization reactor by contacting the hydrogen-enriched liquid feed stream with a solid reverse isomerization catalyst for reverse hydroisomerization in a two-phase liquid-solid reverse isomerization fixed-bed reaction zone under conditions that minimize cracking reactions and that are effective to convert iso-paraffins to normal paraffins and recovering an equilibrium composition of normal and iso paraffins in an isomerization effluent stream; providing the isomerization effluent stream to a stabilization column to separate the isomerization effluent stream into a C.sub.1-C.sub.4 hydrocarbon stream and a stabilized isomerate stream; and combining the stabilized isomerate stream with the light naphtha feedstock to generate the mixed feed stream.
2. The process of claim 1, wherein the hydrocarbon feed stream additionally comprises a second light naphtha feedstock.
3. The process of claim 2, wherein the light naphtha feedstock and the second light naphtha

feedstock comprise the same composition.

4. The process of claim 1, wherein the light naphtha feedstock has a RON of 60 or less.

5. The process of claim 4, wherein the isomerate effluent stream has a RON of at least 80.

6. The process of claim 1, wherein the reverse isomerization reactor is operated at a temperature of from 20° C. to 300° C.

7. The process of claim 1, wherein the reverse isomerization reactor is operated at a pressure of from 10 bars to 100 bars.

8. The process of claim 1, wherein the reverse isomerization reactor is operated at a LHSV of 0.2 to 20 h.sup.-1.

9. The process of claim 1, wherein the hydrogen-enriched liquid feed stream comprises a hydrogen to hydrocarbon mole ratio of 0.01 to 20.0.

10. The process of claim 1, wherein the conditions within the reverse isomerization reactor are effective to maintain at least 90 V % of the hydrogen-enriched liquid feed stream in liquid phase.

11. The process of claim 1, wherein the solid reverse isomerization catalyst comprises 0.05 wt. % to 5 wt. % of at least one Group VIIIB metal.

12. The process of claim 1, wherein the solid reverse isomerization catalyst comprises a base material including zeolite and metal oxides with metals from Group IIIA-B or IVA-B.

13. The process of claim 1, wherein the one or more reverse isomerization reactors comprises multiple individual reactors in series.

14. The process of claim 1, wherein the reverse isomerization reactor is operated at conditions effective to maintain at least 98 V % of the hydrogen-enriched liquid feed stream in liquid phase and include a temperature of from 100° C. to 180° C., a pressure of from 20 bars to 70 bars, and a liquid hourly space velocity of 0.2 to 20 h.sup.-1, with a hydrogen to hydrocarbon mole ratio of 0.01 to 20.0 in the hydrogen-enriched liquid feed stream.

15. The process of claim 1, wherein the one or more normal paraffin streams comprises less than 20 percent by weight of iso-paraffins.

16. The process of claim 1, wherein the separation unit comprises a deisopentanizer column and a fractionator, wherein: the mixed feed stream is provided to the deisopentanizer column to separate iso-pentane as an iso-pentane stream from the mixed feed stream to generate a deisopentanizer effluent comprising normal pentane and hexanes, the deisopentanizer effluent is provided to the fractionator to generate an iso-hexanes stream, a normal pentane stream, and a normal hexane stream; and combining the iso-pentane stream and the iso-hexanes stream to generate the iso-paraffin stream.

17. The process of claim 1, wherein the separation unit comprises a molecular sieve separation column, wherein the molecular sieve separation column separates the mixed feed stream into the iso-paraffin stream and the one or more normal paraffin streams with adsorption based separation.
